Programming Fundamentals

Lab 2

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Roll Number: 2

Program: BS Computer Science (BSCS)

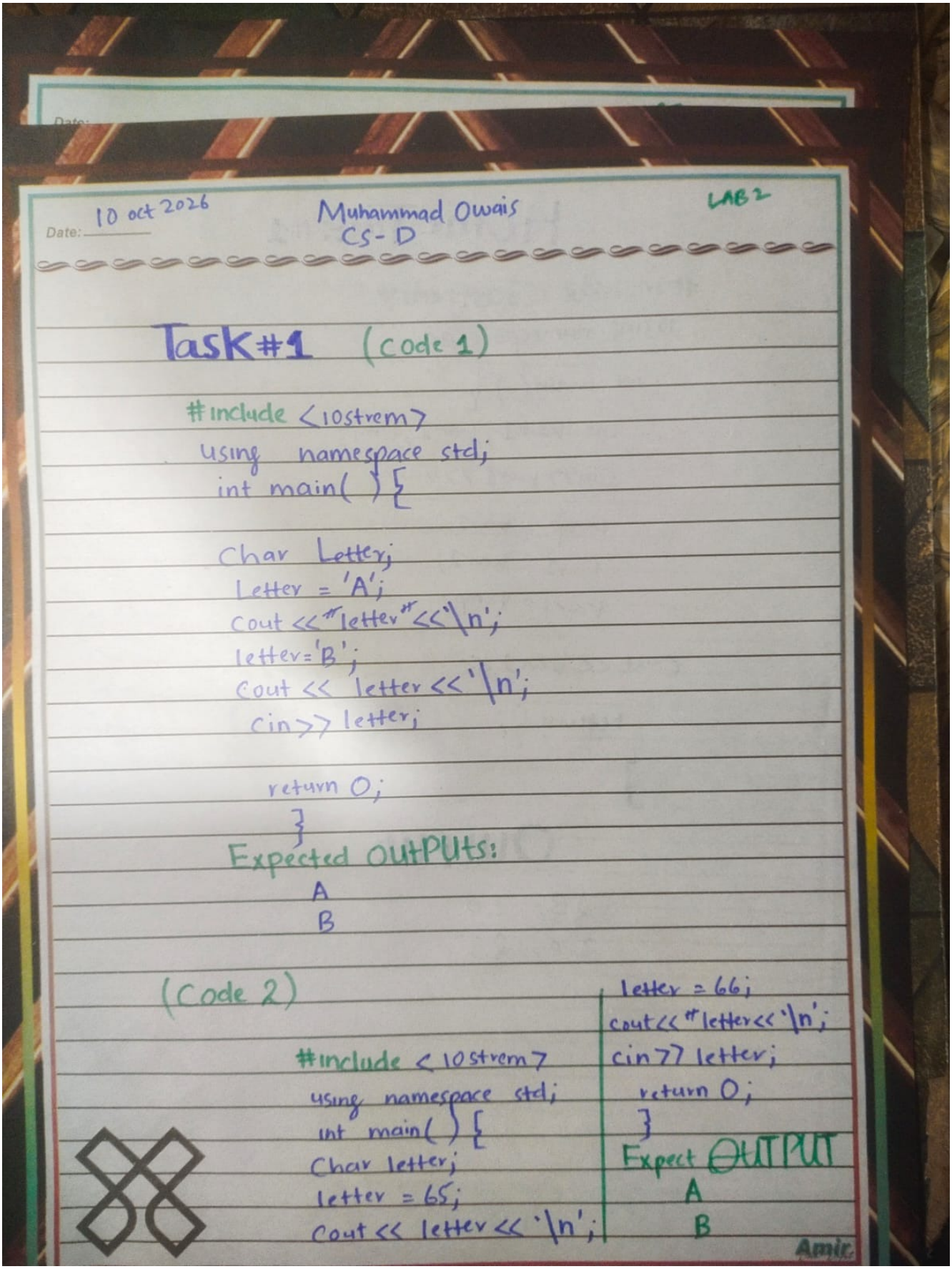
Section: D

University: Ghazi University

Regular Lab Tasks

Task 1 - Code 1

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Program Output

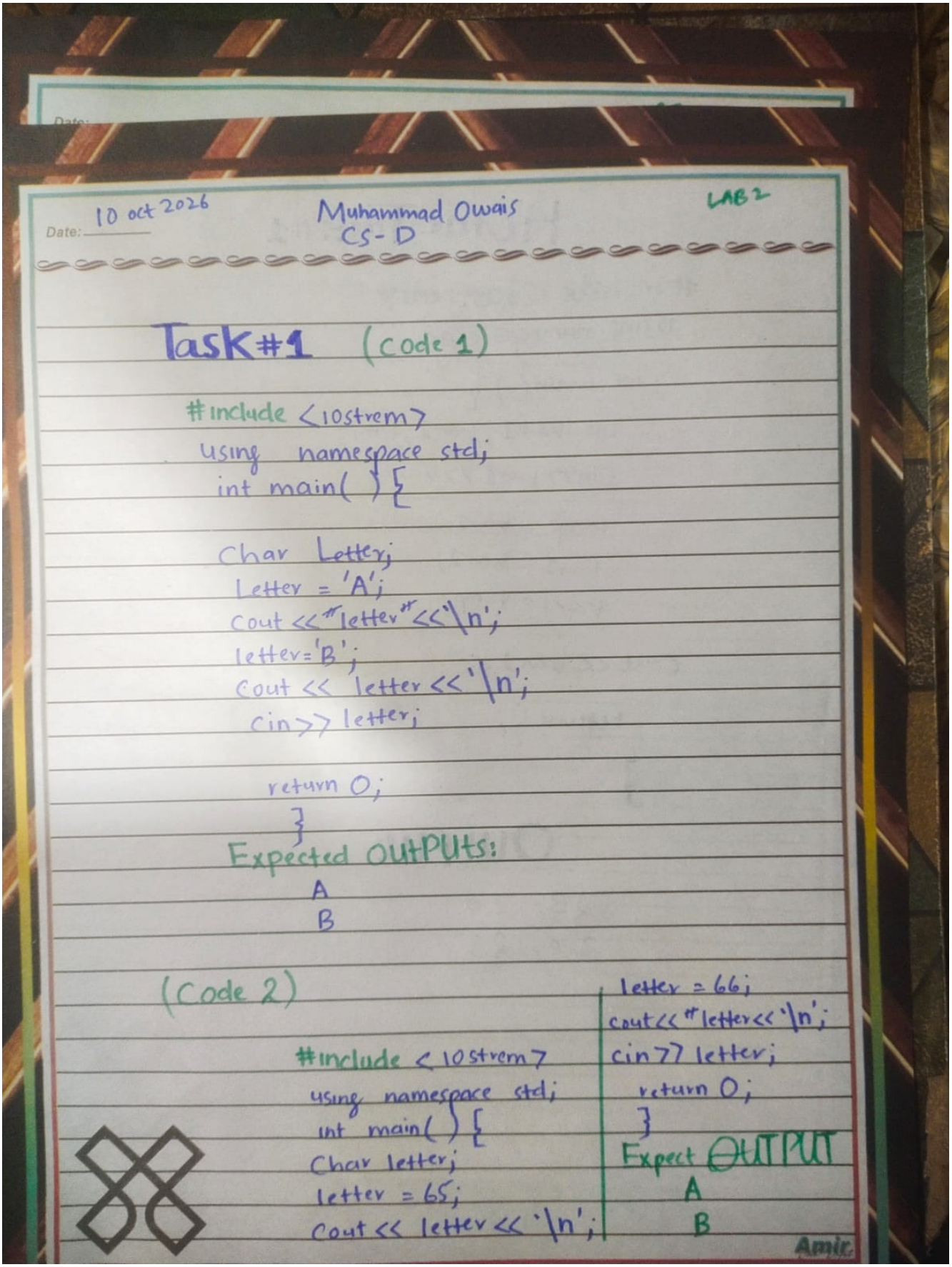
Problems Output Debug Console Terminal Ports

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
A
B
S
```

Note: Codes 1 and 2 of Task 1 share a single handwritten page, which is included in both sections.

Task 1 - Code 2

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Program Output

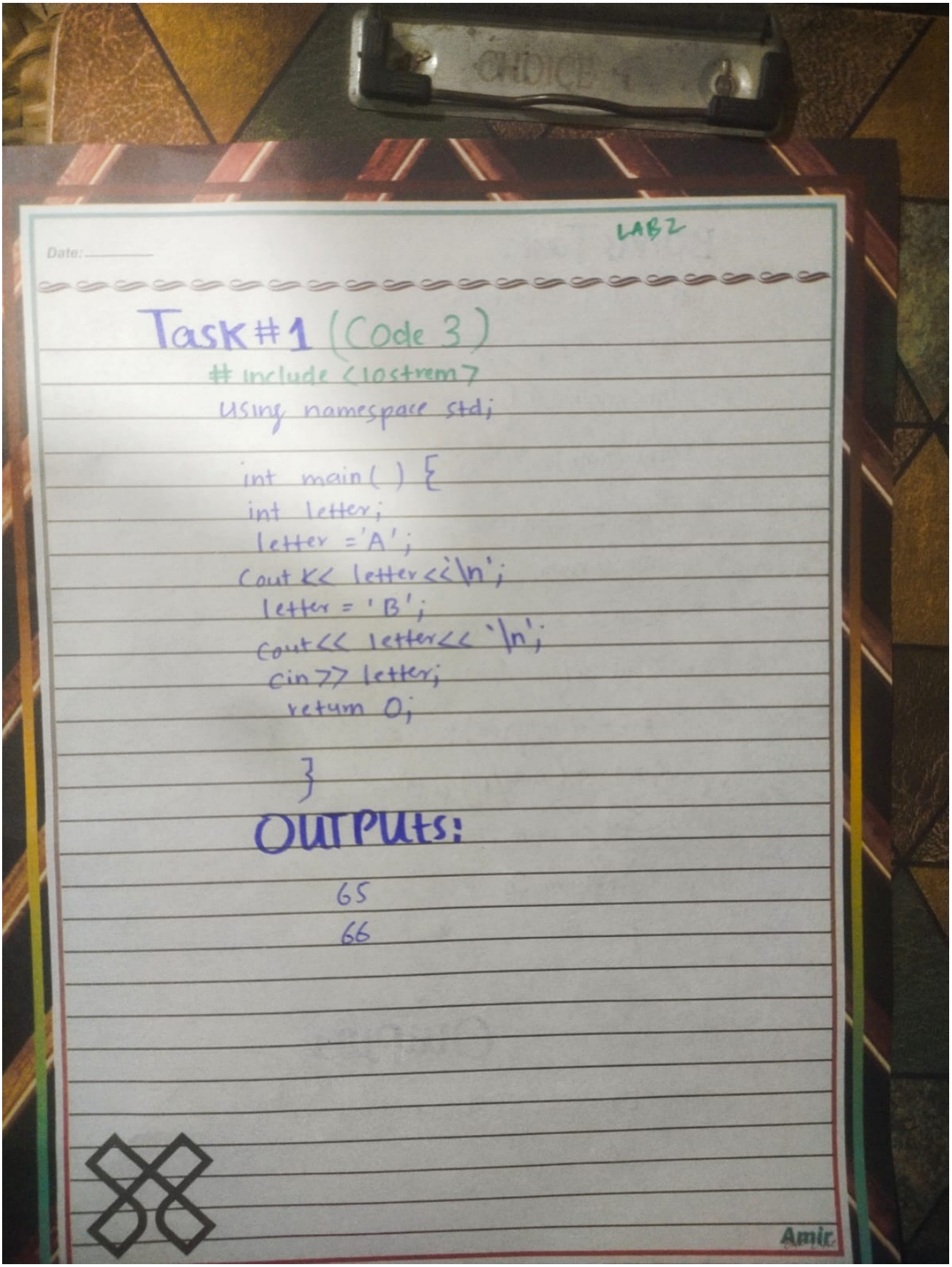
Problems Output Debug Console Terminal Ports

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
A
B
```

powersh... Code Code Code Code C/C++: ... cppdbg: m...

Task 1 - Code 3

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Program Output

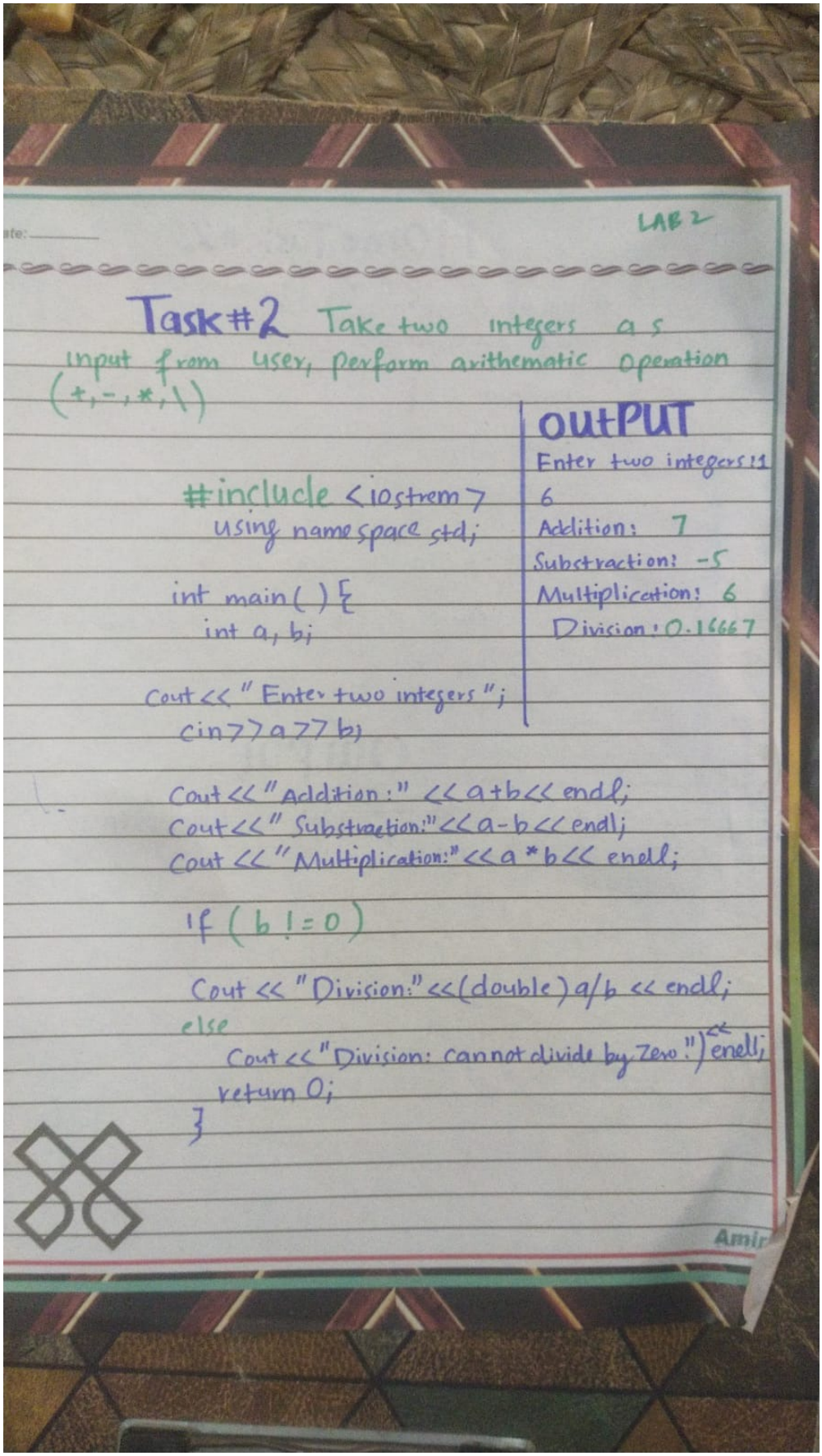
Problems Output Debug Console Terminal Ports

```
    ion
     + FullyQualifiedErrorId : CommandNotFoundException

PS E:\C++>
PS E:\C++>  ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\PC\.vscode\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-jsa2abtx.2yn' '--stdout=Microsoft-MIEngine-Out-dndyflug.b3m' '--stderr=Microsoft-MIEngine-Error-svna5nib.shd' '--pid=Microsoft-MIEngine-Pid-wfonoe0x.b5n' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
65
66
```

Task 2

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;

    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Addition: " << a + b << endl;
    cout << "Subtraction: " << a - b << endl;
    cout << "Multiplication: " << a * b << endl;

    if (b != 0)
        cout << "Division: " << (double)a / b << endl;
    else
        cout << "Division: Cannot divide by zero." << endl;

    return 0;
}
```

Program Output

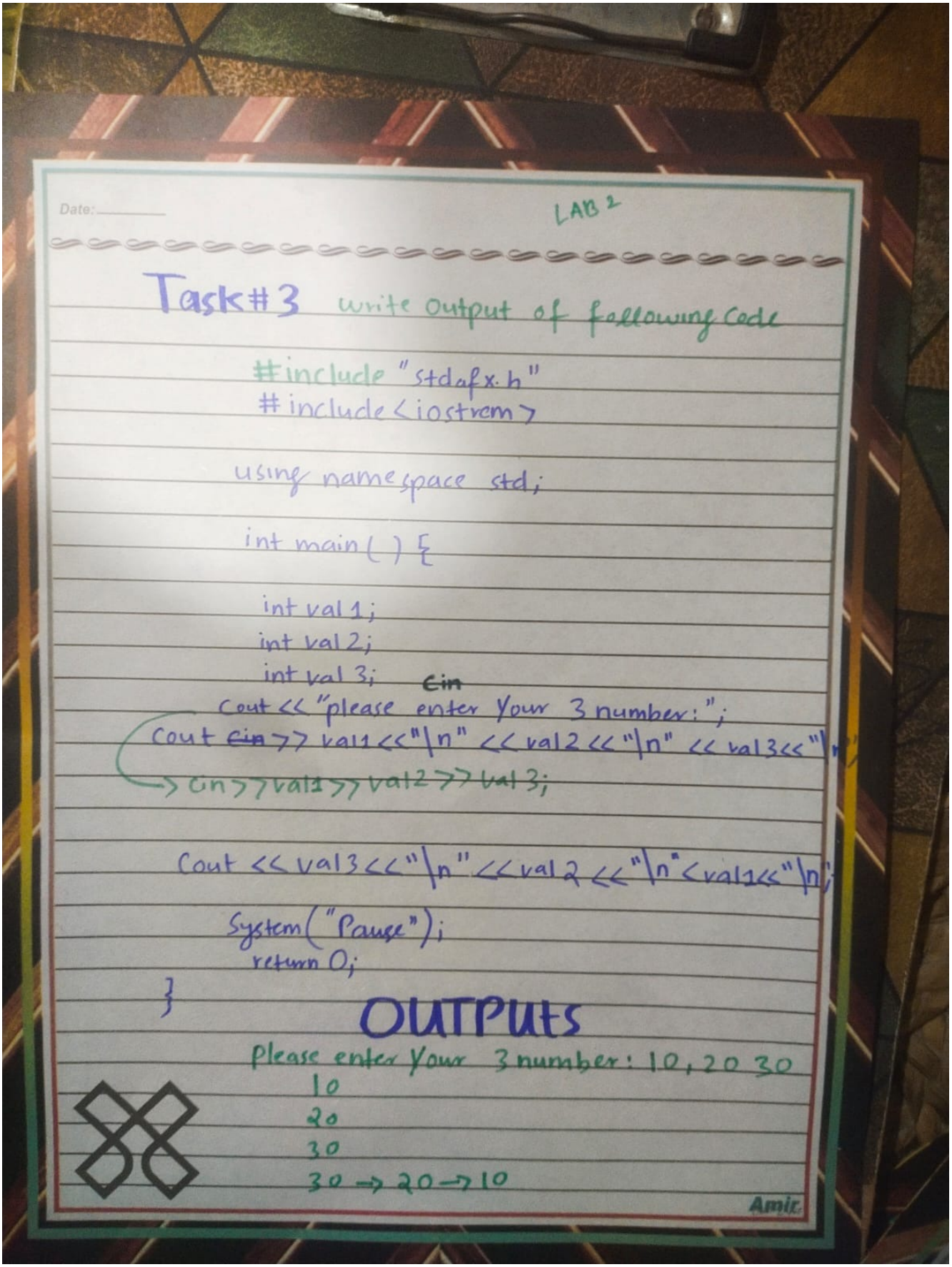
Problems Output Debug Console Terminal Ports

```
Warning: PowerShell detected that you might be using a screen reader and has disable
d PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Mo
dule PSReadLine'.

PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter two integers: 1
6
Addition: 7
Subtraction: -5
Multiplication: 6
Division: 0.166667
PS E:\C++>
```

Task 3

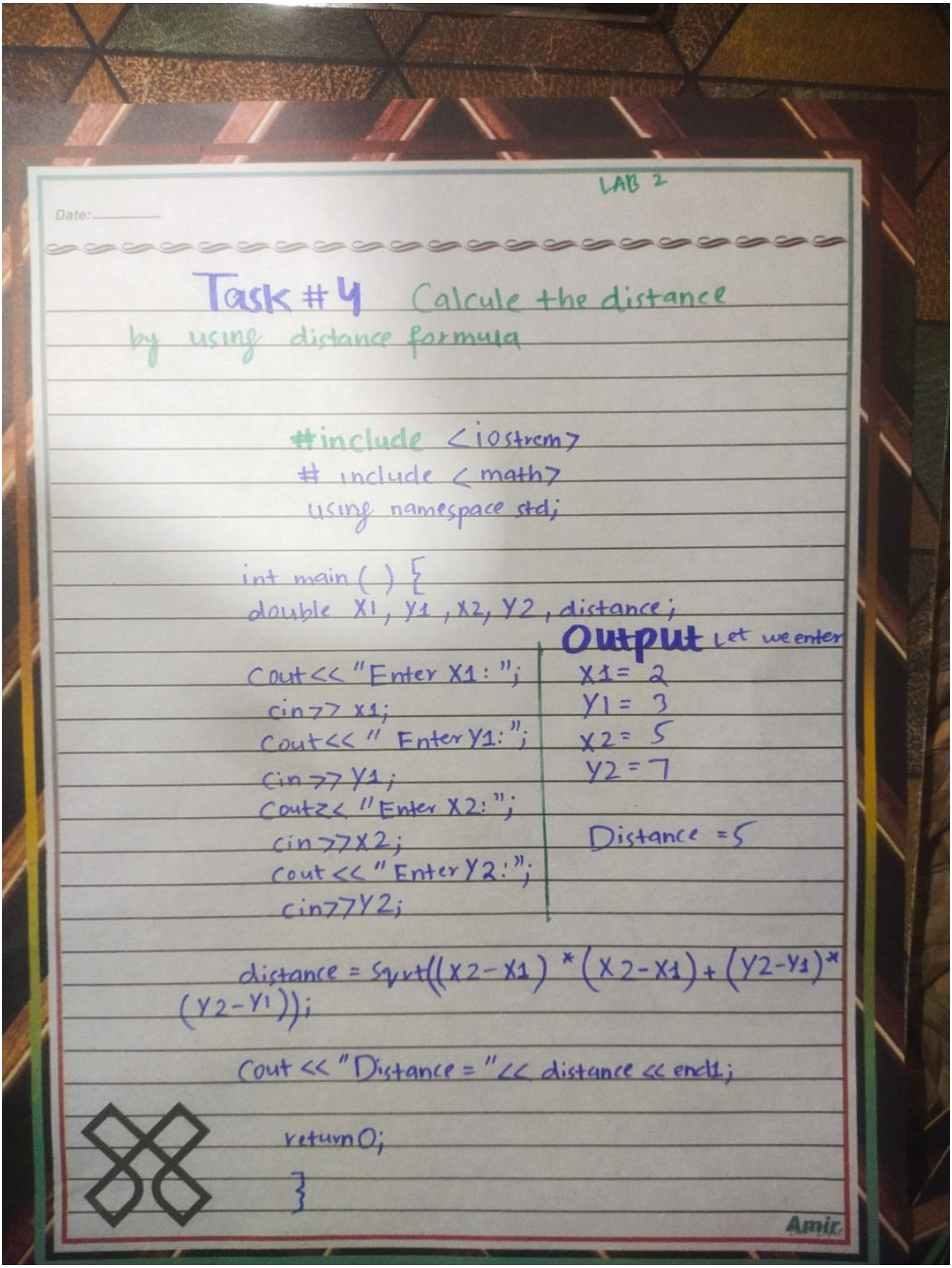
Handwritten Code



Typed-code and program-output screenshots for Task 3 were not included in the uploaded files. The handwritten page above contains the written output of the code.

Task 4

Handwritten Code



Typed Code

main.cpp

```cpp

int main() {
    double x1, y1, x2, y2, distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    distance = sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

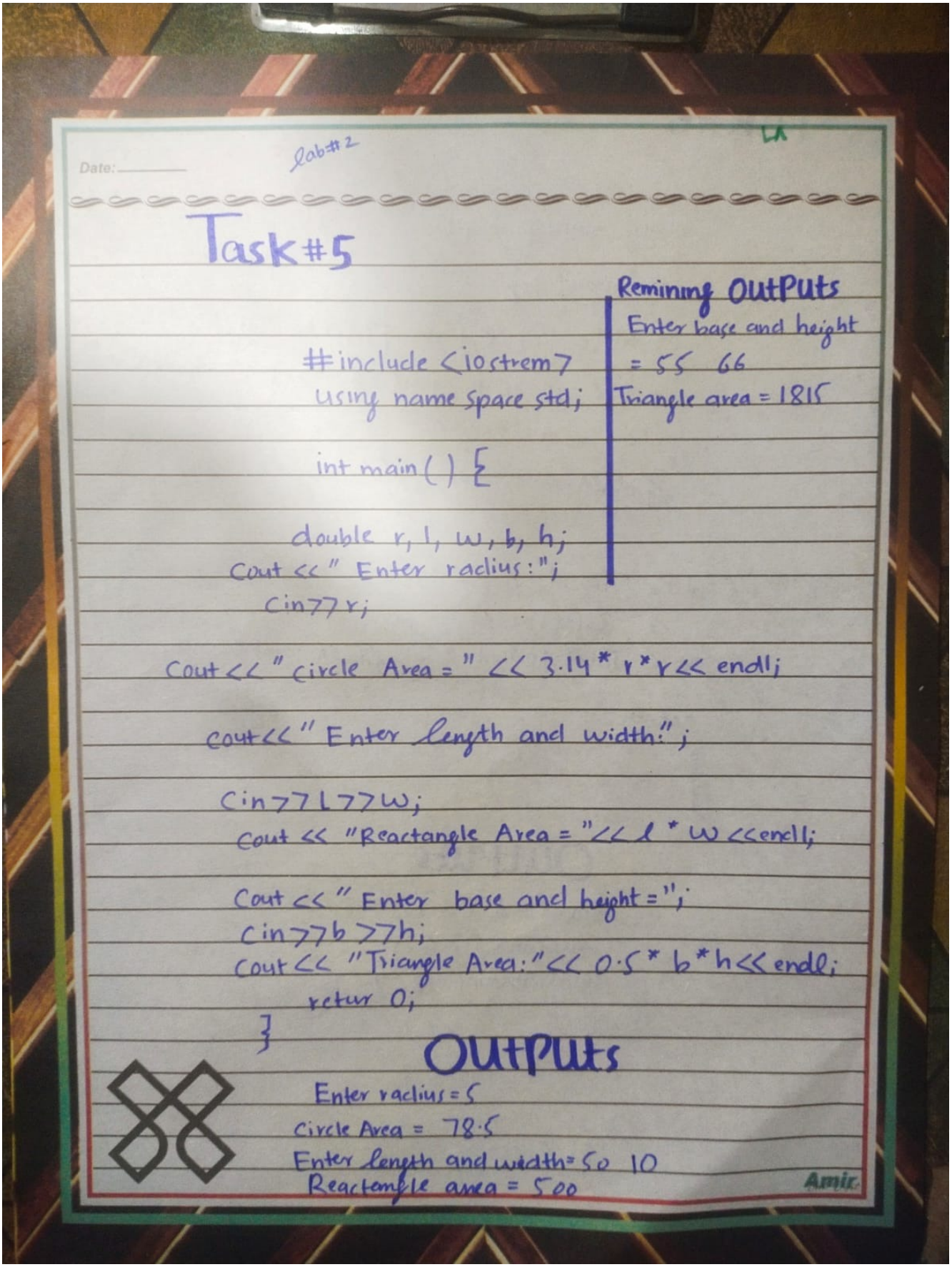
Program Output

Problems Output Debug Console Terminal Ports

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter x1: 2
Enter y1: 3
Enter x2: 5
Enter y2: 7
Distance = 5
PS E:\C++>
```

Task 5

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    double r, l, w, b, h;

    cout << "Enter radius: ";
    cin >> r;
    cout << "Circle Area = " << 3.14 * r * r << endl;

    cout << "Enter length and width: ";
    cin >> l >> w;
    cout << "Rectangle Area = " << l * w << endl;

    cout << "Enter base and height: ";
    cin >> b >> h;
    cout << "Triangle Area = " << 0.5 * b * h << endl;

    return 0;
}
```

Program Output

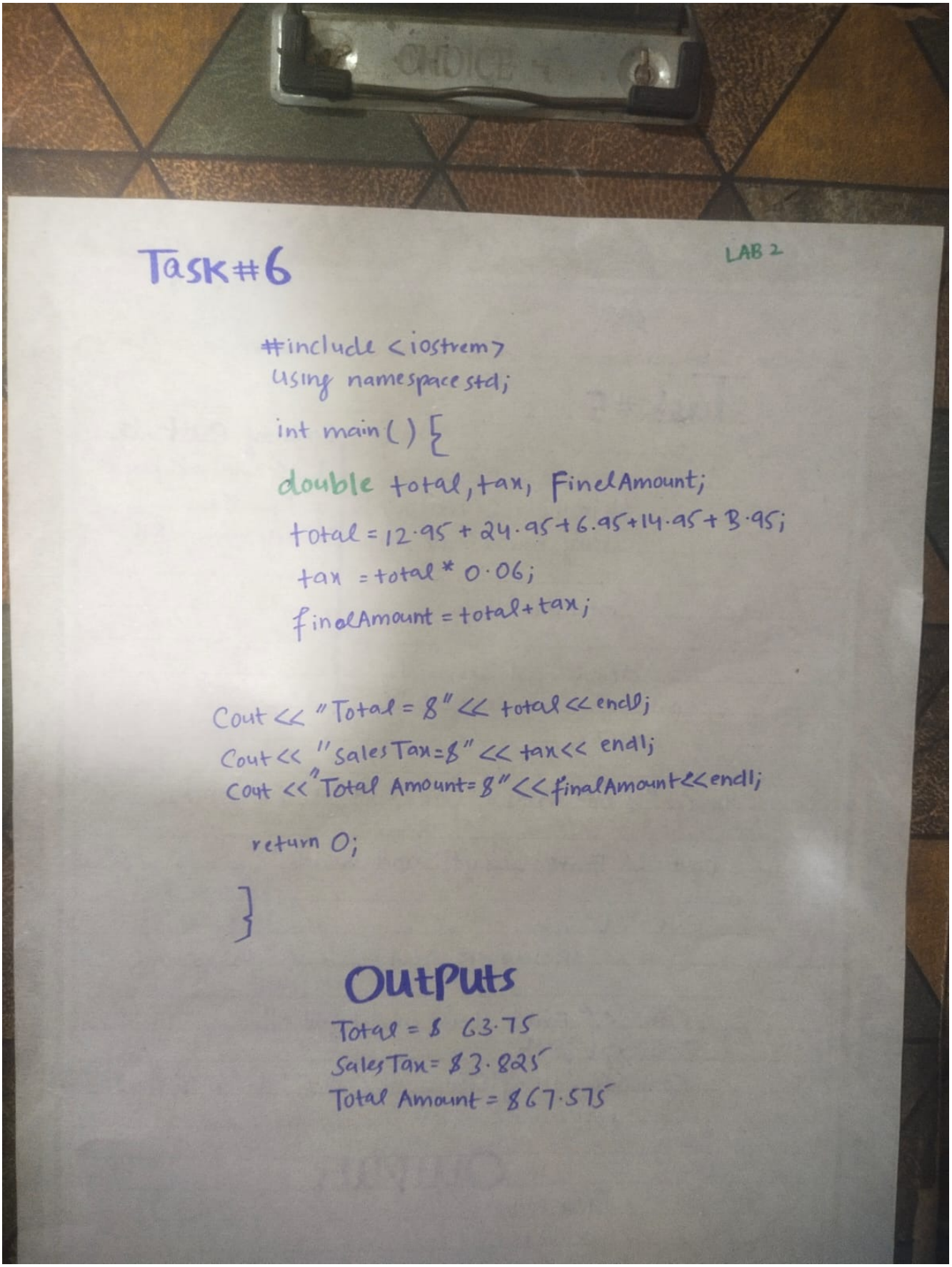
Problems Output Debug Console Terminal Ports Code

```
  compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter radius: 5
Circle Area = 78.5
Enter length and width: 50
10
Rectangle Area = 500
Enter base and height: 55
66
Triangle Area = 1815
PS E:\C++>
```

Task 6

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    double total, tax, finalAmount;

    total = 12.95 + 24.95 + 6.95 + 14.95 + 3.95;
    tax = total * 0.06;
    finalAmount = total + tax;

    cout << "Total = $" << total << endl;
    cout << "Sales Tax = $" << tax << endl;
    cout << "Total Amount = $" << finalAmount << endl;

    return 0;
}
```

Program Output

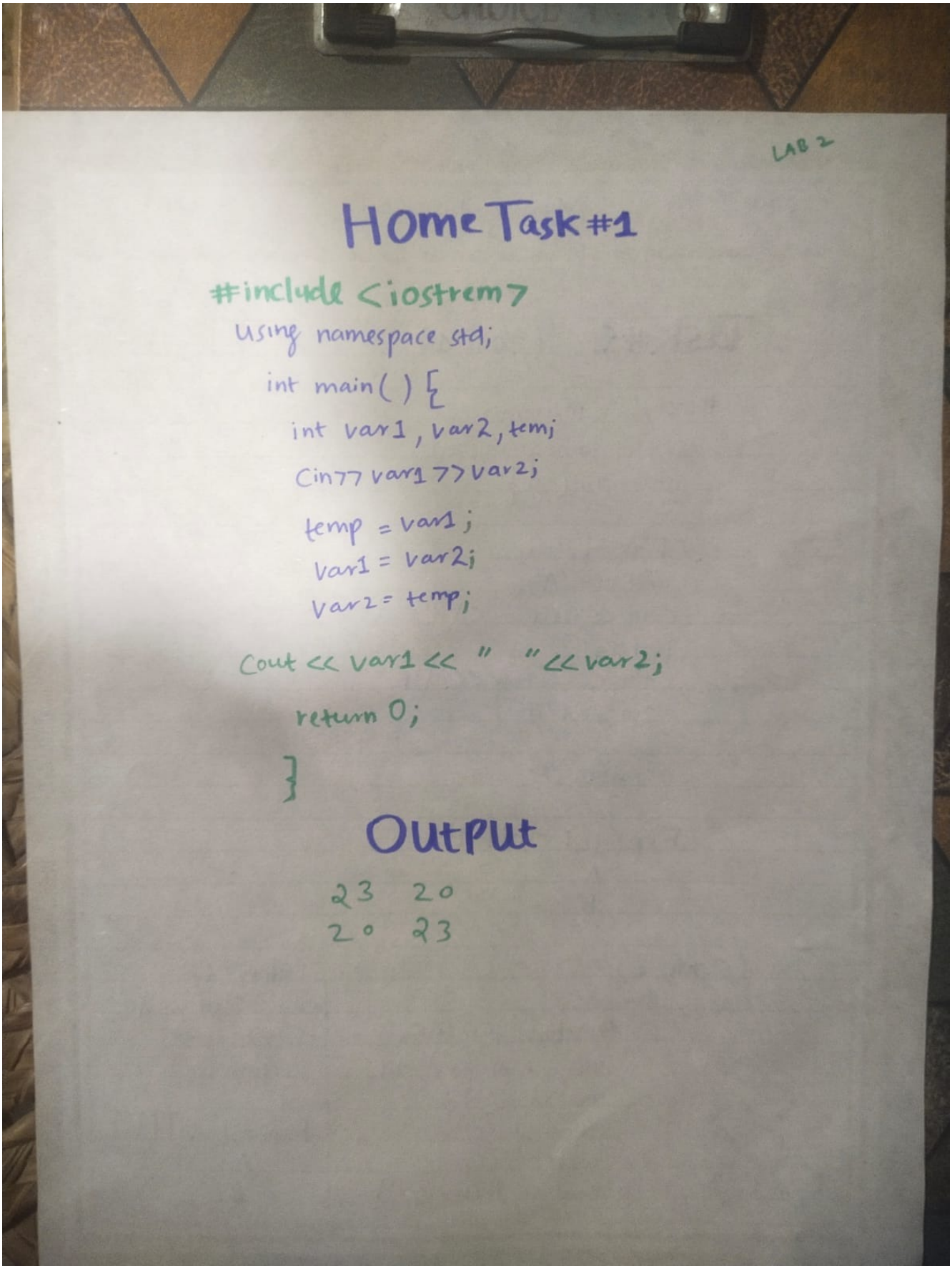
Problems Output Debug Console Terminal Ports

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Total = $63.75
Sales Tax = $3.825
Total Amount = $67.575
PS E:\C++>
```

Home Tasks

Home Task 1

Handwritten Code



Typed Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int var1, var2, temp;
    cin >> var1 >> var2;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << var1 << " " << var2;

    return 0;
}
```

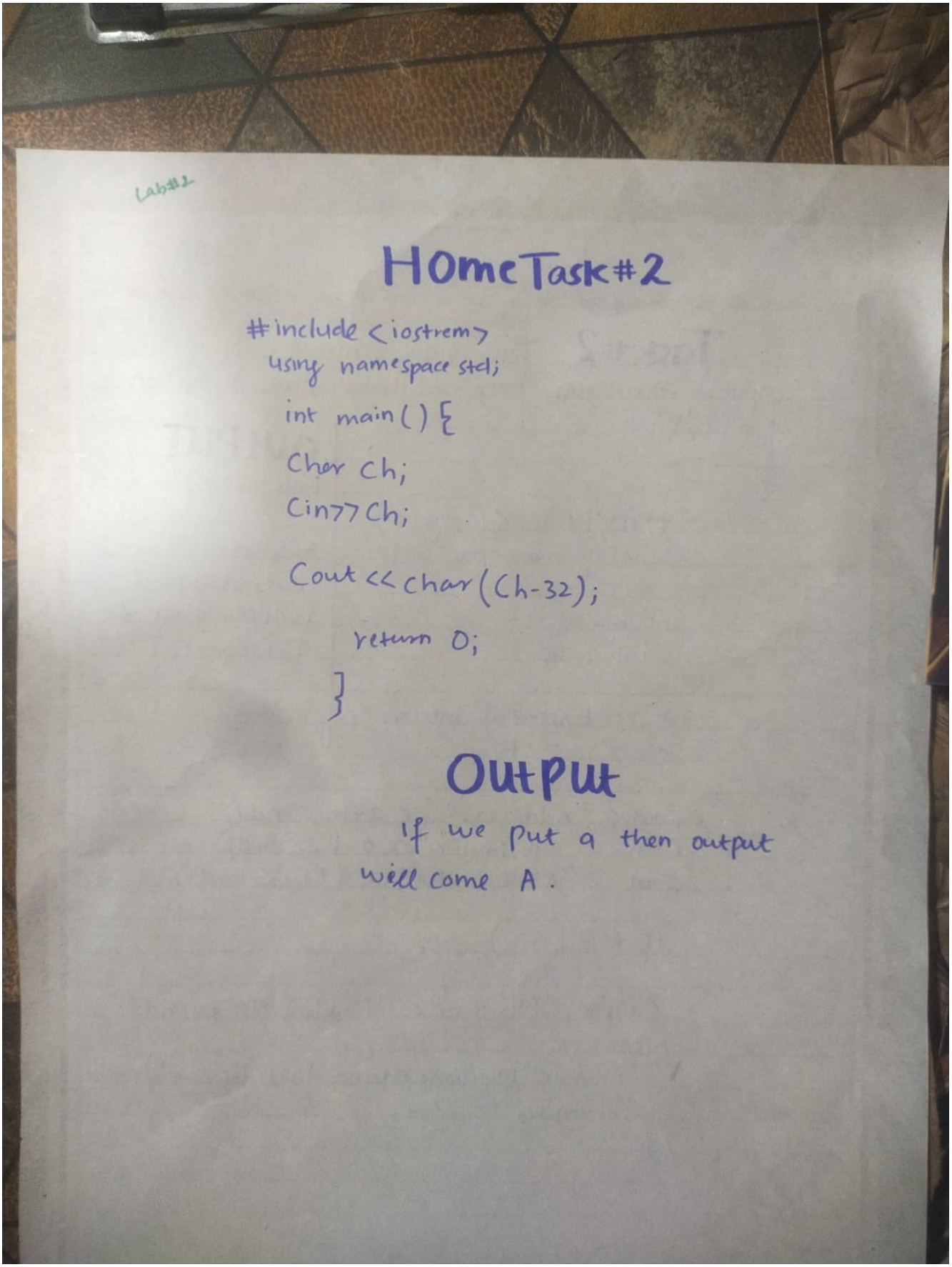
Program Output

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }

23 20
20 23
PS E:\C++>
```

Home Task 2

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    char ch;
    cin >> ch;

    cout << char(ch - 32);

    return 0;
}
```

Program Output

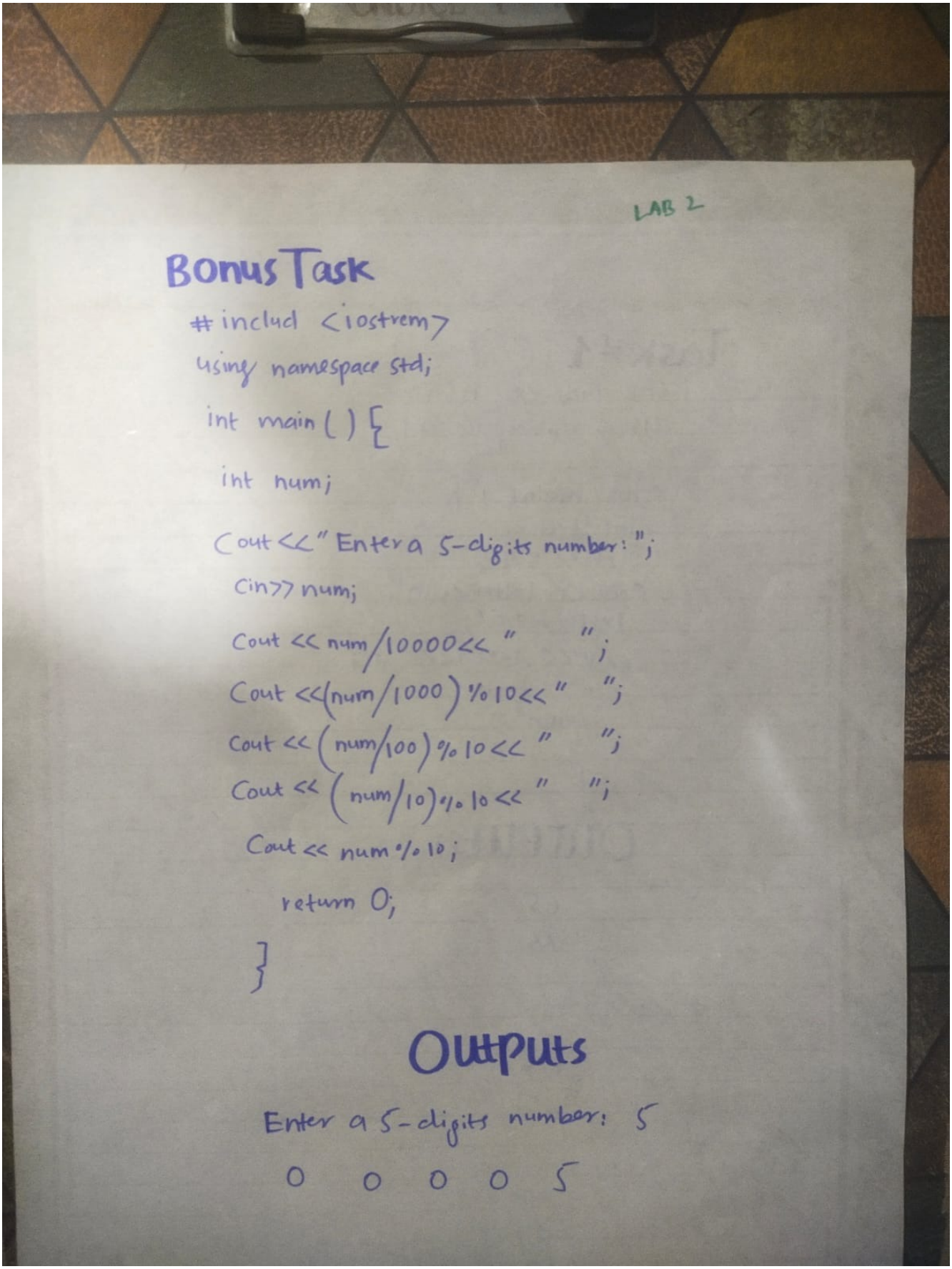
Problems Output Debug Console Terminal Ports Code

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
a
A
PS E:\C++>
```

Bonus Task

Bonus Task

Handwritten Code



Typed Code

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a 5-digit number: ";
    cin >> num;

    cout << num / 10000 << "   ";
    cout << (num / 1000) % 10 << "   ";
    cout << (num / 100) % 10 << "   ";
    cout << (num / 10) % 10 << "   ";
    cout << num % 10;

    return 0;
}
```

Program Output

Problems Output Debug Console Terminal Ports Code

```
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter a 5-digit number: 5
0   0   0   0   5
PS E:\C++>
```